RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 5 (Lambda)
Topic(s): Vectors
(Vectors II and III (Solutions))

1 **(a)** With reference to the origin O, the points A, B and X are $\overrightarrow{OA}=\mathbf{a}$, $\overrightarrow{OB}=\mathbf{b}$ and $\overrightarrow{OX}=\frac{1}{8}\mathbf{a}+\frac{3}{8}\mathbf{b}$. The point Y lies on AB such that O, X and Y are collinear. Express $\overrightarrow{OY}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ and find the ratio of $AY:YB$. [5]

(b) The points P, Q and R have position vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ respectively. P and Q are fixed, and R varies. Describe geometrically the set of possible positions of the point R such that

(i) $(\mathbf{r}-\mathbf{p})\times\mathbf{q}=\mathbf{0}$, [2]

(ii) $(\mathbf{r}-\mathbf{p}).\mathbf{q}=0$. [2]

2 There are three distinct points A, B and P. The point P lies on the circle with centre O and diameter AB. Relative to the origin O, the position vectors of points A and P are $\mathbf{a}$ and $\mathbf{p}$ respectively.

(a) By expressing $\overrightarrow{AP}$ and $\overrightarrow{BP}$ in terms of $\mathbf{a}$ and $\mathbf{p}$, show that $\overrightarrow{AP}.\overrightarrow{BP}=0$. [3]

The point C with position vector $\mathbf{c}$ lies on AB produced such that $AC:AB$ is $\lambda:1$.

(b) Find $\mathbf{c}$ in terms of λ and $\mathbf{a}$. [2]

(c) It is given that PC is a tangent to the circle and that angle AOP is $120°$. Using a suitable scalar product, find the value of λ. [4]

3 Relative to the origin O, points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are non-zero vectors that are not parallel to one another. The points A, B and C are not collinear.

A point of trisection is a point that divides a line segment internally in the ratio $1:2$ or $2:1$. Suppose another two points D and E are points of trisection of line segments AB and AC respectively and both points are nearer to A than to B and C respectively. The lines BE and CD meet at point F.

(i) Show that the vector equations of the lines BE and CD can be expressed as $\mathbf{r}=\frac{2}{3}\lambda\mathbf{a}+(1-\lambda)\mathbf{b}+\frac{1}{3}\lambda\mathbf{c}$ and $\mathbf{r}=\frac{2}{3}\mu\mathbf{a}+\frac{1}{3}\mu\mathbf{b}+(1-\mu)\mathbf{c}$ respectively, where λ and μ are parameters. Hence, show that at point F, $\alpha\mathbf{a}+\beta\mathbf{b}+\gamma\mathbf{c}=\mathbf{0}$, where α, β and γ are constants, each to be expressed in terms of λ and μ. [4]

(ii) Given further that $OACB$ is a parallelogram, find the position vector of F in terms of $\mathbf{a}$ and $\mathbf{b}$. [3]

4 The line l_1 has equation $\mathbf{r} = 11\mathbf{i} - 3\mathbf{j} - \mathbf{k} + \lambda(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$, where λ is a parameter. The point A has coordinates $(2, 0, -1)$.

(a) The plane p contains the line l_1 and the point A. Find a cartesian equation of the plane p.

[3]

(b) Find the position vector of the point A', the reflection of the point A in the line l_1.

[4]

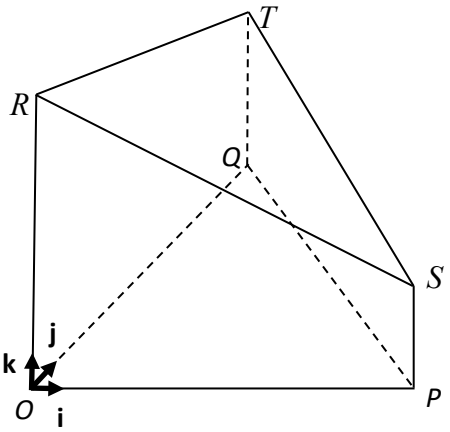
The line l_2 passes through the point A, and is perpendicular to the line l_1. Planes p and q are perpendicular planes such that p and q meet at line l_2.

(c) Find a vector equation of the line l_2.

[2]

(d) The plane Π is such that Π contains the point $(11, -3, -1)$ and is parallel to q. Find the perpendicular distance between Π and q. [2]

5



A tent is constructed with three vertical poles situated at O, P and Q on the horizontal ground. The poles OR, PS and QT have heights 16m, 6m and 8m respectively. It is given that OP = 15m and OQ = 20m. The point O is taken as the origin, with unit vectors **i** along OP, **j** along OQ and **k** along OR.

(i) Find, in the form $ax + by + cz = 240$, the equation of the plane RST. [3]

(ii) Find the angle between the plane RST and the horizontal ground. [2]

(iii) Find the perpendicular distance from P to the plane RST. [2]

(iv) To stabilise the tent, one end of a bar of fixed length 20m is attached to the point R, with the other end of the bar to be attached to a point A on PQ. Find the coordinates of A. [4]

6 A drone flies in a straight line towards a rooftop for inspection. The drone's flight path is modelled by the line $l: \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$ where $\lambda \in \mathbb{R}$, and the rooftop is modelled by the plane $p: 2x + y - z = 7$.

(a) On Day 1, the drone flies towards the rooftop using the flight path. Upon reaching the rooftop, the drone flies in a new flight path, *m*. The acute angle between the original flight path and the normal to the rooftop is the same as the acute angle between the new flight path and the same normal to the rooftop. Both flight paths and the normal lie on the same plane. Find the cosine of the angle between the original flight path and the new flight path in an exact non-trigonometrical form. [3]

(b) On Day 2, the drone starts to descend from point *A* with coordinates $(2,1,0)$ towards the rooftop using the shortest possible path. Find the exact coordinates of the point where the drone lands on the rooftop. [4]

(c) After landing, the drone glides on the rooftop in another new flight path, *s*, that is perpendicular to the flight path *l*. Find a cartesian equation of the flight path *s*. [3]

A second rooftop is built parallel to the first rooftop *p* such that the perpendicular distance between these two rooftops is *d* units. Find two possible cartesian equations of the second rooftop in terms of *d*. [3]

7 The line ℓ has the equation $\frac{x-a}{2} = y-2 = \frac{1-2z}{b}$, where a and b are real constants. The plane π has the equation $3x-y+4z=10$.

(a) Given that the line ℓ and the plane π do not intersect, show that $a \neq \frac{10}{3}$ and $b = \frac{5}{2}$. [5]

(b) It is given that $a=1$ and $b=3$. Find the point of intersection between the line ℓ and the plane π. [3]

(c) It is given now that $a=1$, $b=\frac{5}{2}$.

(i) Find the distance between the line ℓ and the plane π. [2]

(ii) Determine whether the line ℓ and the origin O lie on the same side of the plane π. State an equation of the other plane that is equidistant from line ℓ and parallel to plane π. [3]

8 The line l has vector equation

$$\mathbf{r} = \begin{pmatrix} b \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ a \\ -5 \end{pmatrix},$$

where λ is a real parameter and a, b are real constants.

The equation of plane π_1 is given as $3x-y+2z=7$.

(a) Given that the distance between l and π_1 is non-zero, find the value of a, and the set of values of b. [4]

For the rest of the question, let a be the value found in part **(a)**, and $b=2$.

(b) Find the vector equation of the line l', where l' is the reflection of l in π_1. [4]

The equation of plane π_2 is given as $-3x+y-2z=2$.

(c) Given that the ratio of the distance between π_1 and π_2 to the distance between π_2 and plane π_3 is $2:3$, find the possible cartesian equations of π_3. [4]